Web Programming Laboratory

Experiment 3 - Recitation

Question 1

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>AI Art Studio - Pricing Plans</title> <style> * {
margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif; overflow-x: hidden; } .split-container { display: flex; min-height: 100vh; } .hero-section {
flex: 1; background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); padding: 60px; color: white; display:
flex; flex-direction: column; justify-content: center; } .hero-title { font-size: 4em; margin-bottom: 15px;
text-shadow: 2px 2px 4px rgba(0,0,0,0.2); } .hero-subtitle { font-size: 1.5em; opacity: 0.95; margin-bottom:
40px; } .art-grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 20px; margin-bottom: 30px; }
.art-item { aspect-ratio: 1; border-radius: 15px; overflow: hidden; box-shadow: 0 10px 30px rgba(0,0,0,0.3);

transition: transform 0.3s ease; } .art-item:hover { transform: scale(1.05); } .art-item img { width: 100%;
height: 100%; object-fit: cover; } .art-features { display: flex; gap: 15px; flex-wrap: wrap; }
.feature-badge { background: rgba(255,255,255,0.2); backdrop-filter: blur(10px); padding: 12px 20px;
border-radius: 25px; font-size: 1.05em; } .pricing-section { flex: 1; padding: 60px; background: #f8f9fa;
overflow-y: auto; } .pricing-section h1 { font-size: 3em; color: #2c3e50; margin-bottom: 40px; text-align:
center; } .pricing-container { display: flex; flex-direction: column; gap: 25px; } .pricing-card {
background: white; border-radius: 20px; padding: 35px; box-shadow: 0 5px 20px rgba(0,0,0,0.1); transition:
transform 0.3s ease, box-shadow 0.3s ease; border-left: 5px solid #667eea; } .pricing-card:hover { transform:
translateX(10px); box-shadow: 0 10px 30px rgba(0,0,0,0.15);

} .pricing-card.pro-plan { border-left-color: #f093fb; position: relative; } .pricing-card.pro-plan:before {
content: '■ POPULAR'; position: absolute; top: -15px; right: 30px; background: linear-gradient(135deg,
#f093fb 0%, #f5576c 100%); color: white; padding: 5px 15px; border-radius: 15px; font-size: 0.85em;
font-weight: bold; } .pricing-card.enterprise-plan { border-left-color: #4facfe; } .pricing-card h2 {
font-size: 2em; color: #2c3e50; margin-bottom: 10px; } .price { font-size: 2.5em; color: #667eea;
font-weight: bold; margin-bottom: 20px; } .pricing-card.pro-plan .price { color: #f093fb; }
.pricing-card.enterprise-plan .price { color: #4facfe; } .features { list-style: none; margin-bottom: 25px; }
.features li { padding: 12px 0; border-bottom: 1px solid #ecf0f1; display: flex; align-items: center; gap:
10px; } .features li:before { content: '✓';

color: #27ae60; font-weight: bold; font-size: 1.2em; } .btn { width: 100%; padding: 15px; background:
linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white; border: none; border-radius: 10px;
font-size: 1.1em; font-weight: bold; cursor: pointer; transition: all 0.3s ease; } .btn:hover { transform:
translateY(-3px); box-shadow: 0 10px 20px rgba(102, 126, 234, 0.3); } .pricing-card.pro-plan .btn {
background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%); } .pricing-card.enterprise-plan .btn {
background: linear-gradient(135deg, #4facfe 0%, #00f2fe 100%); } @media (max-width: 1024px) {
.split-container { flex-direction: column; } } </style> </head> <body> <div class="split-container"> <!--
Left Side: Hero/Art Section --> <div class="hero-section"> <div class="hero-content"> <h1
class="hero-title">■ AI Art Studio</h1> <p class="hero-subtitle">Create stunning artwork with the power of
AI</p> <div class="art-showcase"> <div class="art-grid"> <div class="art-item"> <img
src="https://images.unsplash.com/photo-1634017839464-5c339ebe3cb4?w=400&q=80" alt="AI Art 1"> </div> <div
class="art-item"> <img src="https://images.unsplash.com/photo-1618005182384-a83a8bd57fbe?w=400&q=80" alt="AI
Art 2"> </div> <div class="art-item"> <img
src="https://images.unsplash.com/photo-1620641788421-7a1c342ea42e?w=400&q=80" alt="AI Art 3">

</div> <div class="art-item"> <img src="https://images.unsplash.com/photo-1549317661-bd32c8ce0db2?w=400&q=80"
alt="AI Art 4"> </div> </div> <div class="art-features"> <div class="feature-badge">■ Unlimited
Generations</div> <div class="feature-badge">■ AI-Powered Creation</div> <div class="feature-badge">■■
High-Resolution Output</div> </div> </div> </div> </div> <!-- Right Side: Pricing Section --> <div
class="pricing-section"> <h1>■ Pricing Plans</h1> <div class="pricing-container"> <div class="pricing-card
basic-plan"> <h2>Basic</h2> <p class="price">$9<span style="font-size:0.5em">/month</span></p> <ul
class="features"> <li>100 AI Generations/month</li> <li>Standard Resolution (1024x1024)</li> <li>Basic Styles
& Filters</li> <li>Email Support</li> <li>Personal Use Only</li> </ul> <button class="btn">Get
Started</button> </div> <div class="pricing-card pro-plan"> <h2>Professional</h2> <p class="price">$29<span
style="font-size:0.5em">/month</span></p> <ul class="features"> <li>1000 AI Generations/month</li> <li>High
Resolution (2048x2048)</li> <li>All Styles & Advanced Filters</li> <li>Priority Support</li> <li>Commercial
Use License</li> <li>Custom Training Models</li> </ul> <button class="btn">Choose Pro</button> </div> <div
class="pricing-card enterprise-plan"> <h2>Enterprise</h2> <p class="price">$99<span
style="font-size:0.5em">/month</span></p> <ul class="features"> <li>Unlimited AI Generations</li> <li>Ultra
HD Resolution (4096x4096)</li> <li>All Features + Custom Styles</li> <li>24/7 Dedicated Support</li>

<li>Full Commercial License</li> <li>Private Training Models</li> <li>API Access</li> <li>Team Collaboration
Tools</li> </ul> <button class="btn">Contact Sales</button> </div> </div> </div> </div> </body> </html>
```

Output

🎨 AI Art Studio

Create stunning artwork with the power of AI

🎨 Unlimited Generations

✨ AI-Powered Creation

🖼️ High-Resolution Output

💎 Pricing Plans

Basic

$9/month

- ✓ 100 AI Generations/month
- ✓ Standard Resolution (1024x1024)
- ✓ Basic Styles & Filters
- ✓ Email Support
- ✓ Personal Use Only

Get Started

⭐ POPULAR

Professional

$29/month

- ✓ 1000 AI Generations/month
- ✓ High Resolution (2048x2048)
- ✓ All Styles & Advanced Filters
- ✓ Priority Support
- ✓ Commercial Use License
- ✓ Custom Training Models

Choose Pro

Enterprise

$99/month

- ✓ Unlimited AI Generations
- ✓ Ultra HD Resolution (4096x4096)
- ✓ All Features + Custom Styles
- ✓ 24/7 Dedicated Support
- ✓ Full Commercial License
- ✓ Private Training Models
- ✓ API Access
- ✓ Team Collaboration Tools

Contact Sales

Question 2

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>Poker Hand - Playing Cards</title> <style> * {
margin: 0; padding: 0; box-sizing: border-box; } body { background: linear-gradient(135deg, #0F2027 0%,
#203A43 50%, #2C5364 100%); font-family: 'Arial', sans-serif; padding: 50px 20px; min-height: 100vh; display:
flex; flex-direction: column; align-items: center; justify-content: center; } h1 { color: white; font-size:
3.5em; text-align: center; margin-bottom: 20px; text-shadow: 2px 2px 4px rgba(0,0,0,0.5); } .subtitle {
color: rgba(255,255,255,0.8); font-size: 1.3em; text-align: center; margin-bottom: 50px; } #playing-cards {
display: flex; gap: 20px; flex-wrap: wrap; justify-content: center; perspective: 1000px; } .card { width:
180px; height: 250px; background: white; border-radius: 15px; padding: 20px; display: flex; flex-direction:
column;
```

```
justify-content: space-between; box-shadow: 0 15px 40px rgba(0,0,0,0.5); transition: all 0.3s ease; cursor:
pointer; position: relative; } .card:hover { transform: translateY(-20px) rotateY(10deg); box-shadow: 0 25px
60px rgba(0,0,0,0.6); } .card:before { content: ''; position: absolute; top: 0; left: 0; right: 0; bottom: 0;
background: linear-gradient(135deg, transparent 0%, rgba(255,255,255,0.1) 100%); border-radius: 15px;
pointer-events: none; } .card .left, .card .right { display: flex; flex-direction: column; align-items:
center; gap: 5px; } .card .left { align-self: flex-start; } .card .right { align-self: flex-end; transform:
rotate(180deg); } .card .middle { flex: 1; display: flex; align-items: center; justify-content: center; }
.rank { font-size: 2em; font-weight: bold; } .suit { font-size: 1.8em; } .suit-large { font-size: 6em;
```

```
} .hearts .rank, .hearts .suit, .hearts .suit-large { color: #e74c3c; } .diamonds .rank, .diamonds .suit,
.diamonds .suit-large { color: #e74c3c; } .spades .rank, .spades .suit, .spades .suit-large { color: #2c3e50;
} .clubs .rank, .clubs .suit, .clubs .suit-large { color: #2c3e50; } .hand-info { text-align: center;
margin-top: 40px; background: rgba(255,255,255,0.1); backdrop-filter: blur(10px); padding: 20px 40px;
border-radius: 15px; color: white; } .hand-info h2 { font-size: 2em; margin-bottom: 10px; color: #ffd700; }
.hand-info p { font-size: 1.2em; opacity: 0.9; } @media (max-width: 768px) { #playing-cards { gap: 15px; }
.card { width: 140px; height: 200px; padding: 15px; } .suit-large { font-size: 4em !important;
```

```
} } </style> </head> <body> <h1>■ Poker Hand</h1> <p class="subtitle">Royal Flush - The Best Hand in
Poker!</p> <div id="playing-cards"> <!-- Ace of Hearts --> <div class="card hearts"> <div class="left"> <div
class="rank">A</div> <div class="suit">♥</div> </div> <div class="middle"> <div class="suit-large">♥</div>
</div> <div class="right"> <div class="rank">A</div> <div class="suit">♥</div> </div> </div> <!-- King of
Hearts --> <div class="card hearts"> <div class="left"> <div class="rank">K</div> <div class="suit">♥</div>
</div> <div class="middle"> <div class="suit-large">♥</div> </div> <div class="right"> <div
class="rank">K</div> <div class="suit">♥</div> </div> </div> <!-- Queen of Hearts --> <div class="card
hearts"> <div class="left"> <div class="rank">Q</div> <div class="suit">♥</div> </div> <div class="middle">
<div class="suit-large">♥</div> </div> <div class="right"> <div class="rank">Q</div> <div
class="suit">♥</div> </div> </div>
```

```
<!-- Jack of Hearts --> <div class="card hearts"> <div class="left"> <div class="rank">J</div> <div
class="suit">♥</div> </div> <div class="middle"> <div class="suit-large">♥</div> </div> <div class="right">
<div class="rank">J</div> <div class="suit">♥</div> </div> </div> <!-- 10 of Hearts --> <div class="card
hearts"> <div class="left"> <div class="rank">10</div> <div class="suit">♥</div> </div> <div class="middle">
<div class="suit-large">♥</div> </div> <div class="right"> <div class="rank">10</div> <div
class="suit">♥</div> </div> </div> </div> <div class="hand-info"> <h2>■ Royal Flush!</h2> <p>A♥ K♥ Q♥ J♥ 10♥
- The highest ranking hand in poker</p> </div> </body> </html>
```

Output

Poker Hand
Royal Flush - The Best Hand in Poker!
A
K
Q
J
10
Royal Flush!
A♥ K♥ Q♥ J♥ 10♥ - The highest ranking hand in poker

Question 3

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>C++ Documentation</title> <style> * { margin: 0;
padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
background: #f5f7fa; display: flex; min-height: 100vh; } #navbar { width: 300px; background:
linear-gradient(180deg, #1e3c72 0%, #2a5298 100%); color: white; position: fixed; height: 100vh; overflow-y:
auto; box-shadow: 3px 0 10px rgba(0,0,0,0.1); } #navbar header { font-size: 2em; padding: 30px 20px;
border-bottom: 2px solid rgba(255,255,255,0.2); font-weight: bold; } .nav-link { display: block; color:
rgba(255,255,255,0.9); text-decoration: none; padding: 15px 20px; border-left: 4px solid transparent;
transition: all 0.3s ease; } .nav-link:hover { background: rgba(255,255,255,0.1); border-left-color: #4fc3f7;
padding-left: 25px; } #main-doc { margin-left: 300px; padding: 50px; flex: 1;
```

```
} .main-section { background: white; border-radius: 15px; padding: 40px; margin-bottom: 30px; box-shadow: 0
5px 20px rgba(0,0,0,0.08); } .main-section header { font-size: 2.5em; color: #1e3c72; margin-bottom: 20px;
border-bottom: 3px solid #4fc3f7; padding-bottom: 10px; } .main-section p { line-height: 1.8; color: #34495e;
margin-bottom: 15px; font-size: 1.05em; } code { display: block; background: #2c3e50; color: #4fc3f7;
padding: 20px; border-radius: 8px; margin: 20px 0; font-family: 'Courier New', monospace; overflow-x: auto;
font-size: 0.95em; line-height: 1.6; } ul, ol { margin: 15px 0 15px 30px; line-height: 2; color: #34495e; } ul
li, ol li { margin: 8px 0; } @media (max-width: 768px) { #navbar { width: 100%; position: relative; height:
auto; } #main-doc { margin-left: 0; padding: 20px;
```

```
} } </style> </head> <body> <!-- Navigation Bar --> <nav id="navbar"> <header>■ C++ Docs</header> <a
href="#Introduction" class="nav-link">Introduction</a> <a href="#Getting_Started" class="nav-link">Getting
Started</a> <a href="#Variables_and_Data_Types" class="nav-link">Variables and Data Types</a> <a
href="#Control_Flow" class="nav-link">Control Flow</a> <a href="#Functions" class="nav-link">Functions</a> <a
href="#Classes_and_Objects" class="nav-link">Classes and Objects</a> </nav> <!-- Main Documentation --> <main
id="main-doc"> <!-- Section 1: Introduction --> <section class="main-section" id="Introduction">
<header>Introduction</header> <p>C++ is a powerful, high-performance programming language developed by Bjarne
Stroustrup at Bell Labs in 1979 as an extension of the C programming language. It combines the efficiency and
control of low-level programming with high-level features like object-oriented programming.</p> <p>C++ is one
of the most widely used programming languages in the world, particularly in systems programming, game
development, embedded systems, and performance-critical applications where speed and resource efficiency are
crucial.</p> <p>The language supports multiple programming paradigms including procedural, object-oriented,
and generic programming. C++ gives programmers fine-grained control over system resources and memory, making
it ideal for applications where performance is critical.</p> <code>#include &lt;iostream&gt; int main() {
std::cout &lt;&lt; "Hello, World!" &lt;&lt; std::endl; return 0; }</code> <p>This classic example
demonstrates a basic C++ program. The language is used in operating systems, web browsers, databases, game
engines, and many other applications that require high performance and efficient resource management.</p>
</section> <!-- Section 2: Getting Started --> <section class="main-section" id="Getting_Started">
<header>Getting Started</header> <p>To start programming in C++, you need a C++ compiler. Popular choices
include GCC (GNU Compiler Collection), Clang, and Microsoft Visual C++. Many integrated development
environments (IDEs) come with compilers built-in.</p> <p>To compile a C++ program from the command line using
GCC, save your code in a file with a .cpp extension and use:</p> <code>g++ program.cpp -o program
./program</code> <p>There are several ways to write and compile C++ code:</p> <ul> <li>Command-line compilers
like GCC or Clang</li> <li>IDEs such as Visual Studio, CLion, Code::Blocks, or Eclipse</li> <li>Online
compilers for quick testing and learning</li> <li>Text editors with build system integration like VS Code or
Sublime Text</li> </ul> </section> <!-- Section 3: Variables and Data Types --> <section class="main-section"
id="Variables_and_Data_Types"> <header>Variables and Data Types</header> <p>In C++, every variable must be
declared with a specific data type. The type determines what kind of values the variable can hold and how much
memory it will use.</p> <p>Common fundamental data types in C++ include:</p>
```

```
<ul> <li><strong>int:</strong> Integer numbers (e.g., 42, -17, 0)</li> <li><strong>double:</strong>
Floating-point numbers (e.g., 3.14, -0.5)</li> <li><strong>char:</strong> Single characters (e.g., 'A', '7',
'$')</li> <li><strong>bool:</strong> Boolean values (true or false)</li> <li><strong>string:</strong>
Sequence of characters (requires #include &lt;string&gt;)</li> </ul> <code>int age = 25; double pi = 3.14159;
char grade = 'A'; bool isPassed = true; std::string name = "John Doe";</code> </section> <!-- Section 4:
Control Flow --> <section class="main-section" id="Control_Flow"> <header>Control Flow</header> <p>Control
flow statements allow you to control the execution path of your program based on conditions or loops.</p>
<p><strong>If-Else Statements:</strong></p> <code>if (score >= 90) { std::cout &lt;&lt; "Grade: A" &lt;&lt;
std::endl; } else if (score >= 80) { std::cout &lt;&lt; "Grade: B" &lt;&lt; std::endl; } else { std::cout
&lt;&lt; "Grade: C" &lt;&lt; std::endl; }</code> <p><strong>For Loops:</strong></p> <code>for (int i = 0; i
&lt; 10; i++) { std::cout &lt;&lt; i &lt;&lt; " "; }</code> <p><strong>While Loops:</strong></p> <code>int
count = 0; while (count &lt; 5) { std::cout &lt;&lt; count &lt;&lt; std::endl; count++; }</code> </section>
<!-- Section 5: Functions --> <section class="main-section" id="Functions"> <header>Functions</header>
<p>Functions are reusable blocks of code that perform specific tasks. They help organize code, reduce
repetition, and improve maintainability.</p> <p>A function declaration includes a return type, name,
parameters, and body:</p> <code>int add(int a, int b) { return a + b; } int main() { int result = add(5, 3);
std::cout &lt;&lt; "Result: " &lt;&lt; result &lt;&lt; std::endl; return 0; }</code> <p>Functions can have
```

```
different return types:</p>
```

```
<ul> <li><strong>void:</strong> Returns nothing</li> <li><strong>int, double, etc.:</strong> Returns a value
of that type</li> <li>Functions can also accept parameters by value or by reference</li> </ul> </section>
<!-- Section 6: Classes and Objects --> <section class="main-section" id="Classes_and_Objects">
<header>Classes and Objects</header> <p>C++ is an object-oriented language. Classes are blueprints for
creating objects. They encapsulate data and functions that operate on that data.</p> <code>class Rectangle {
private: int width; int height; public: Rectangle(int w, int h) { width = w; height = h; } int area() { return
width * height; } }; int main() { Rectangle rect(10, 5); std::cout &lt;&lt; "Area: " &lt;&lt; rect.area()
&lt;&lt; std::endl; return 0; }</code> <p>Key OOP concepts in C++:</p> <ul>
<li><strong>Encapsulation:</strong> Bundling data and methods together</li> <li><strong>Inheritance:</strong>
Creating new classes from existing ones</li> <li><strong>Polymorphism:</strong> Objects of different types
responding to the same interface</li> <li><strong>Abstraction:</strong> Hiding complex implementation
details</li> </ul> </section> </main> </body> </html>
```

Output

C++ Docs

- Introduction
- Getting Started
- Variables and Data Types
- Control Flow
- Functions
- Classes and Objects

Introduction

C++ is a powerful, high-performance programming language developed by Bjarne Stroustrup at Bell Labs in 1979 as an extension of the C programming language. It combines the efficiency and control of low-level programming with high-level features like object-oriented programming.

C++ is one of the most widely used programming languages in the world, particularly in systems programming, game development, embedded systems, and performance-critical applications where speed and resource efficiency are crucial.

The language supports multiple programming paradigms including procedural, object-oriented, and generic programming. C++ gives programmers fine-grained control over system resources and memory, making it ideal for applications where performance is critical.

```
#include <iostream> int main() { std::cout << "Hello, World!" << std::endl; return 0; }
```

This classic example demonstrates a basic C++ program. The language is used in operating systems, web browsers, databases, game engines, and many other applications that require high performance and efficient resource management.

Getting Started

To start programming in C++, you need a C++ compiler. Popular choices include GCC (GNU Compiler Collection), Clang, and Microsoft Visual C++. Many integrated development environments (IDEs) come with compilers built-in.

To compile a C++ program from the command line using GCC, save your code in a file with a .cpp extension and use:

```
g++ program.cpp -o program ./program
```

There are several ways to write and compile C++ code:

- Command-line compilers like GCC or Clang
- IDEs such as Visual Studio, CLion, Code::Blocks, or Eclipse
- Online compilers for quick testing and learning
- Text editors with build system integration like VS Code or Sublime Text

Variables and Data Types

In C++, every variable must be declared with a specific data type. The type determines what kind of values the variable can hold and how much memory it will use.

Common fundamental data types in C++ include:

- **int:** Integer numbers (e.g., 42, -17, 0)
- **double:** Floating-point numbers (e.g., 3.14, -0.5)
- **char:** Single characters (e.g., 'A', '7', '$')
- **bool:** Boolean values (true or false)
- **string:** Sequence of characters (requires #include <string>)

```
int age = 25; double pi = 3.14159; char grade = 'A'; bool isPassed = true; std::string name = "John Doe";
```

Control Flow

Control flow statements allow you to control the execution path of your program based on conditions or loops.

If-Else Statements:

```
if (score >= 90) { std::cout << "Grade: A" << std::endl; } else if (score >= 80) { std::cout << "Grade: B"
<< std::endl; } else { std::cout << "Grade: C" << std::endl; }
```

For Loops:

```
for (int i = 0; i < 10; i++) { std::cout << i << " "; }
```

While Loops:

```
int count = 0; while (count < 5) { std::cout << count << std::endl; count++; }
```

Functions

Functions are reusable blocks of code that perform specific tasks. They help organize code, reduce repetition, and improve maintainability.

A function declaration includes a return type, name, parameters, and body:

```
int add(int a, int b) { return a + b; } int main() { int result = add(5, 3); std::cout << "Result: " <<
result << std::endl; return 0; }
```

Functions can have different return types:

- **void:** Returns nothing
- **int, double, etc.:** Returns a value of that type
- Functions can also accept parameters by value or by reference

Classes and Objects

C++ is an object-oriented language. Classes are blueprints for creating objects. They encapsulate data and functions that operate on that data.

```
class Rectangle { private: int width; int height; public: Rectangle(int w, int h) { width = w; height = h; }
int area() { return width * height; } }; int main() { Rectangle rect(10, 5); std::cout << "Area: " <<
rect.area() << std::endl; return 0; }
```

Key OOP concepts in C++:

- **Encapsulation:** Bundling data and methods together
- **Inheritance:** Creating new classes from existing ones
- **Polymorphism:** Objects of different types responding to the same interface
- **Abstraction:** Hiding complex implementation details

Question 4

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>The Daily Tech Tribune</title> <style> * { margin: 0;
padding: 0; box-sizing: border-box; } body { font-family: 'Georgia', serif; background: #f5f5f5; padding:
30px; } .newspaper-layout { max-width: 1200px; margin: 0 auto; background: white; padding: 40px; box-shadow:
0 10px 40px rgba(0,0,0,0.1); } .title { text-align: center; border-bottom: 4px double #2c3e50;
padding-bottom: 20px; margin-bottom: 30px; } .title h1 { font-size: 4em; font-family: 'Times New Roman',
serif; color: #2c3e50; letter-spacing: 2px; margin-bottom: 10px; } .subtitle { color: #7f8c8d; font-size:
1.1em; font-style: italic; } article h2 { font-size: 2em; color: #2c3e50; margin-bottom: 15px; line-height:
1.3; } article p { line-height: 1.8;
```

```
color: #34495e; margin-bottom: 15px; text-align: justify; font-size: 1.05em; } .feature-article {
column-count: 2; column-gap: 40px; margin-bottom: 30px; padding-bottom: 30px; border-bottom: 2px solid
#ecf0f1; } .feature-article h2 { column-span: all; font-size: 2.5em; color: #e74c3c; } .cover-image { width:
100%; margin: 30px 0; } .cover-image img { width: 100%; height: 400px; object-fit: cover; border-radius: 8px;
} .cover-image figcaption { text-align: center; font-style: italic; color: #7f8c8d; margin-top: 10px;
font-size: 0.95em; } .secondary-article { margin: 30px 0; padding: 30px 0; border-bottom: 1px solid #ecf0f1;
} .secondary-article h2 { color: #3498db; } .articles-grid { display: grid; grid-template-columns: repeat(2,
1fr); gap: 30px; margin-top: 30px; } .small-article { padding: 20px;
```

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background: #f8f9fa; border-left: 4px solid #27ae60; border-radius: 5px; } .small-article h3 { font-size:
1.5em; color: #2c3e50; margin-bottom: 10px; } .small-article p { font-size: 1em; } .quote-box { background:
#ecf0f1; border-left: 5px solid #e74c3c; padding: 20px; margin: 20px 0; font-style: italic; font-size: 1.2em;
color: #2c3e50; } @media (max-width: 768px) { .feature-article { column-count: 1; } .articles-grid {
grid-template-columns: 1fr; } .title h1 { font-size: 2.5em; } } </style> </head> <body> <main
class="newspaper-layout"> <!-- Header/Title --> <header class="title"> <h1>■ The Daily Tech Tribune</h1> <p
class="subtitle">Breaking News in Technology | Thursday, March 5, 2026</p> </header> <!-- Feature Article -->
<article class="feature-article"> <h2>AI Revolution Transforms Global Industries</h2> <p>Artificial
Intelligence continues to reshape the landscape of modern business and technology. Companies worldwide are
reporting unprecedented efficiency gains through the implementation of advanced machine learning systems.
Industry experts predict that AI-driven automation will create millions of new jobs while transforming
traditional roles across all sectors.</p> <p>The impact extends beyond business, with healthcare, education,
and scientific research seeing remarkable breakthroughs. Machine learning algorithms are now capable of
diagnosing diseases with accuracy rivaling human specialists, while educational AI tutors provide
personalized learning experiences to students globally.</p> <p>"We're witnessing a fundamental shift in how
humans interact with technology," says Dr. Sarah Chen, leading AI researcher at Stanford University. "The
integration of AI into everyday life is no longer science fiction—it's our present reality."</p> <p>Major
corporations have invested over $500 billion in AI research and development over the past year alone. This
massive investment has yielded remarkable results, from natural language processing systems that understand
context and nuance to computer vision algorithms that surpass human accuracy in medical imaging analysis.</p>
</article> <!-- Cover Image -->
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<figure class="cover-image"> <img
src="https://images.unsplash.com/photo-1677442136019-21780ecad995?w=1200&q=80" alt="AI Technology">
<figcaption>AI-powered systems are revolutionizing industries worldwide</figcaption> </figure> <!-- Quote Box
--> <div class="quote-box"> "We're witnessing a fundamental shift in how humans interact with technology. The
integration of AI into everyday life is no longer science fiction—it's our present reality." <br><strong>—
Dr. Sarah Chen, Stanford University</strong> </div> <!-- Secondary Article --> <article
class="secondary-article"> <h2>Quantum Computing Reaches New Milestone</h2> <p>Scientists at MIT have
announced a breakthrough in quantum computing stability, maintaining quantum coherence for a record-breaking
10 minutes. This achievement brings practical quantum computers significantly closer to reality, with
potential applications in cryptography, drug discovery, and climate modeling.</p> <p>The research team
utilized a novel approach to error correction, effectively neutralizing one of quantum computing's biggest
challenges. "This is a game-changer," explains lead researcher Dr. James Morrison. "We're moving from
theoretical possibility to practical implementation much faster than anticipated."</p> <p>Major tech
companies including Google, IBM, and Microsoft have already expressed interest in collaborating on the next
phase of development. Industry analysts predict commercial quantum computers could be available within the
next five years, potentially revolutionizing fields from pharmaceutical research to financial modeling.</p>
</article> <!-- Small Articles Grid --> <div class="articles-grid"> <article class="small-article"> <h3>■
SpaceX Launches Record Mission</h3> <p>SpaceX successfully launched its 50th mission of the year, setting a
new industry record. The mission deployed 60 satellites for the expanding Starlink network, bringing
high-speed internet to remote regions worldwide.</p> <p>CEO Elon Musk announced plans for weekly launches
throughout 2026, with ambitious goals for Mars exploration beginning in 2028.</p> </article> <article
class="small-article"> <h3>■ Breakthrough in Cancer Treatment</h3> <p>Researchers have developed a new
immunotherapy treatment showing 85% success rate in clinical trials for previously untreatable cancers. The
treatment harnesses the body's immune system to target cancer cells with unprecedented precision.</p> <p>FDA
approval is expected by the end of the year, offering new hope to millions of patients worldwide.</p>
</article> <article class="small-article"> <h3>■ Climate Tech Investments Soar</h3> <p>Global investments in
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climate technology reached $150 billion in Q1 2026, marking a 300% increase from last year. Renewable energy,
carbon capture, and sustainable agriculture lead the funding surge.</p> <p>Analysts predict climate tech will
become the fastest-growing sector by 2030.</p> </article> <article class="small-article"> <h3>■ 5G Coverage
Reaches 80% Globally</h3> <p>Telecommunications companies report that 5G network coverage now reaches 80% of
the global population. The milestone marks a new era in connectivity, enabling IoT, autonomous vehicles, and
smart cities at scale.</p> <p>6G research has already begun, promising speeds 100x faster than current 5G
networks.</p> </article> </div> </main> </body> </html>
```

Output

📰 The Daily Tech Tribune

Breaking News in Technology | Thursday, March 5, 2026

AI Revolution Transforms Global Industries

Artificial Intelligence continues to reshape the landscape of modern business and technology. Companies worldwide are reporting unprecedented efficiency gains through the implementation of advanced machine learning systems. Industry experts predict that AI-driven automation will create millions of new jobs while transforming traditional roles across all sectors.

The impact extends beyond business, with healthcare, education, and scientific research seeing remarkable breakthroughs. Machine learning algorithms are now capable of diagnosing diseases with accuracy rivaling human specialists, while educational AI tutors provide personalized learning experiences to students globally.

"We're witnessing a fundamental shift in how humans interact with technology," says Dr. Sarah Chen, leading AI researcher at Stanford University. "The integration of AI into everyday life is no longer science fiction—it's our present reality."

Major corporations have invested over $500 billion in AI research and development over the past year alone. This massive investment has yielded remarkable results, from natural language processing systems that understand context and nuance to computer vision algorithms that surpass human accuracy in medical imaging analysis.

AI-powered systems are revolutionizing industries worldwide

> *"We're witnessing a fundamental shift in how humans interact with technology. The integration of AI into everyday life is no longer science fiction—it's our present reality."*
> ***— Dr. Sarah Chen, Stanford University***

Quantum Computing Reaches New Milestone

Scientists at MIT have announced a breakthrough in quantum computing stability, maintaining quantum coherence for a record-breaking 10 minutes. This achievement brings practical quantum computers significantly closer to reality, with potential applications in cryptography, drug discovery, and climate modeling.

The research team utilized a novel approach to error correction, effectively neutralizing one of quantum computing's biggest challenges. "This is a game-changer," explains lead researcher Dr. James Morrison. "We're moving from theoretical possibility to practical implementation much faster than anticipated."

Major tech companies including Google, IBM, and Microsoft have already expressed interest in collaborating on the next phase of development. Industry analysts predict commercial quantum computers could be available within the next five years, potentially revolutionizing fields from pharmaceutical research to financial modeling.

🚀 SpaceX Launches Record Mission

SpaceX successfully launched its 50th mission of the year, setting a new industry record. The mission deployed 60 satellites for the expanding Starlink network, bringing high-speed internet to remote regions worldwide.

CEO Elon Musk announced plans for weekly launches throughout 2026, with ambitious goals for Mars exploration beginning in 2028.

💊 Breakthrough in Cancer Treatment

Researchers have developed a new immunotherapy treatment showing 85% success rate in clinical trials for previously untreatable cancers. The treatment harnesses the body's immune system to target cancer cells with unprecedented precision.

FDA approval is expected by the end of the year, offering new hope to millions of patients worldwide.

🌍 Climate Tech Investments Soar

Global investments in climate technology reached $150 billion in Q1 2026, marking a 300% increase from last year. Renewable energy, carbon capture, and sustainable agriculture lead the funding surge.

Analysts predict climate tech will become the fastest-growing sector by 2030.

📱 5G Coverage Reaches 80% Globally

Telecommunications companies report that 5G network coverage now reaches 80% of the global population. The milestone marks a new era in connectivity, enabling IoT, autonomous vehicles, and smart cities at scale.

6G research has already begun, promising speeds 100x faster than current 5G networks.

Question 5

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>TechWatch Pro - Your Smart Companion</title> <style>
* { margin: 0; padding: 0; box-sizing: border-box; } body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif; background: #fff; } #header { background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
color: white; padding: 20px 50px; position: sticky; top: 0; z-index: 1000; box-shadow: 0 2px 10px
rgba(0,0,0,0.1); } .header-container { max-width: 1200px; margin: 0 auto; display: flex; justify-content:
space-between; align-items: center; } #header-img { height: 50px; } #nav-bar { display: flex; gap: 30px; }
.nav-link { color: white; text-decoration: none; font-size: 1.1em; transition: all 0.3s ease; padding: 8px
16px; border-radius: 5px; } .nav-link:hover { background: rgba(255,255,255,0.2); }
```

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#hero { background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white; text-align: center;
padding: 100px 20px; } #hero h1 { font-size: 4em; margin-bottom: 20px; text-shadow: 2px 2px 4px
rgba(0,0,0,0.2); } #hero p { font-size: 1.5em; margin-bottom: 40px; opacity: 0.95; } #form { display: flex;
gap: 15px; justify-content: center; max-width: 600px; margin: 0 auto; } #email { flex: 1; padding: 15px 20px;
border: none; border-radius: 8px; font-size: 1.1em; } #submit { padding: 15px 40px; background: #27ae60;
color: white; border: none; border-radius: 8px; font-size: 1.1em; font-weight: bold; cursor: pointer;
transition: all 0.3s ease; } #submit:hover { background: #229954; transform: translateY(-2px); box-shadow: 0
5px 15px rgba(39, 174, 96, 0.4); } #features, #how-it-works, #pricing { max-width: 1200px; margin: 80px auto;
padding: 0 20px;
```

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} section h2 { text-align: center; font-size: 3em; color: #2c3e50; margin-bottom: 50px; } .features-grid {
display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 40px; } .feature-item {
text-align: center; padding: 30px; border-radius: 15px; background: #f8f9fa; transition: transform 0.3s ease,
box-shadow 0.3s ease; } .feature-item:hover { transform: translateY(-10px); box-shadow: 0 10px 30px
rgba(0,0,0,0.1); } .feature-icon { font-size: 4em; margin-bottom: 20px; } .feature-item h3 { font-size:
1.8em; color: #2c3e50; margin-bottom: 15px; } .feature-item p { color: #7f8c8d; line-height: 1.6; font-size:
1.1em; } #video { width: 100%; max-width: 800px; height: 450px; display: block; margin: 0 auto;
border-radius: 15px; box-shadow: 0 10px 40px rgba(0,0,0,0.2); } .pricing-grid { display: grid;
grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 30px;
```

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margin-top: 50px; } .pricing-card { background: white; border-radius: 20px; padding: 40px; text-align:
center; box-shadow: 0 10px 40px rgba(0,0,0,0.1); transition: transform 0.3s ease; border: 2px solid #ecf0f1;
} .pricing-card:hover { transform: translateY(-10px); border-color: #667eea; } .pricing-card.featured {
border-color: #667eea; background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white;
transform: scale(1.05); } .pricing-card h3 { font-size: 2em; margin-bottom: 20px; } .price { font-size: 3em;
font-weight: bold; margin: 20px 0; } .pricing-card ul { list-style: none; margin: 30px 0; } .pricing-card li {
padding: 10px 0; border-bottom: 1px solid rgba(0,0,0,0.1); } .pricing-card.featured li { border-bottom-color:
rgba(255,255,255,0.2); } .pricing-btn { padding: 15px 40px; background: #667eea; color: white; border: none;
border-radius: 8px; font-size: 1.1em; cursor: pointer; transition: all 0.3s ease;
```

```
} .pricing-btn:hover { background: #764ba2; transform: translateY(-2px); } .pricing-card.featured
.pricing-btn { background: white; color: #667eea; } @media (max-width: 768px) { #hero h1 { font-size: 2.5em;
} #form { flex-direction: column; } } </style> </head> <body> <!-- Header with Navigation --> <header
id="header"> <div class="header-container"> <img id="header-img"
src="https://via.placeholder.com/50/ffffff/667eea?text=■" alt="TechWatch Pro Logo"> <nav id="nav-bar"> <a
href="#features" class="nav-link">Features</a> <a href="#how-it-works" class="nav-link">How It Works</a> <a
href="#pricing" class="nav-link">Pricing</a> </nav> </div> </header> <!-- Hero Section --> <section
id="hero"> <h1>■ Elevate Your Lifestyle</h1> <p>Experience the future of wearable technology with TechWatch
Pro</p> <form id="form" action="#" method="POST"> <input type="email" id="email" name="email"
placeholder="Enter your email address" required> <input type="submit" id="submit" value="Get Early Access">
</form> </section> <!-- Features Section --> <section id="features"> <h2>Premium Features</h2> <div
class="features-grid"> <div class="feature-item"> <div class="feature-icon">❤■</div> <h3>Health
Monitoring</h3> <p>Track your heart rate, blood oxygen, sleep quality, and stress levels with medical-grade
sensors. Get personalized health insights powered by AI.</p> </div> <div class="feature-item">
```

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<div class="feature-icon">■</div> <h3>Fitness Tracking</h3> <p>Monitor 100+ workout types with automatic
detection. GPS tracking, pace analysis, and real-time coaching to help you reach your fitness goals.</p>
</div> <div class="feature-item"> <div class="feature-icon">■</div> <h3>Water Resistant</h3> <p>50-meter
water resistance means you can swim, surf, or shower with confidence. Built to withstand any adventure.</p>
</div> <div class="feature-item"> <div class="feature-icon">■</div> <h3>Long Battery Life</h3> <p>Up to 7
days on a single charge with typical use. Fast charging gives you a full day's power in just 30 minutes.</p>
</div> <div class="feature-item"> <div class="feature-icon">■</div> <h3>Smart Notifications</h3> <p>Stay
connected with calls, texts, and app alerts on your wrist. Voice assistant integration for hands-free
control.</p> </div> <div class="feature-item"> <div class="feature-icon">■</div> <h3>Customizable Design</h3>
<p>Choose from hundreds of watch faces and interchangeable bands. Express your style with premium materials
and finishes.</p> </div> </div> </section> <!-- How It Works Section --> <section id="how-it-works"> <h2>See
It In Action</h2> <iframe id="video" src="https://www.youtube.com/embed/dQw4w9WgXcQ" frameborder="0"
allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture"
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allowfullscreen> </iframe> </section> <!-- Pricing Section --> <section id="pricing"> <h2>Choose Your
Plan</h2> <div class="pricing-grid"> <div class="pricing-card"> <h3>Starter</h3> <div
class="price">$299</div> <ul> <li>✓ Basic Health Monitoring</li> <li>✓ 50+ Workout Modes</li> <li>✓ 5 Days
Battery</li> <li>✓ Water Resistant</li> <li>✓ 1 Year Warranty</li>
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</ul> <button class="pricing-btn">Buy Now</button> </div> <div class="pricing-card featured"> <h3>■ Pro</h3>
<div class="price">$449</div> <ul> <li>✓ Advanced Health Monitoring</li> <li>✓ 100+ Workout Modes</li> <li>✓
7 Days Battery</li> <li>✓ GPS + Cellular</li> <li>✓ Premium Bands Included</li> <li>✓ 2 Years Warranty</li>
</ul> <button class="pricing-btn">Buy Now</button> </div> <div class="pricing-card"> <h3>Ultra</h3> <div
class="price">$599</div> <ul> <li>✓ Medical-Grade Sensors</li> <li>✓ Unlimited Workouts</li> <li>✓ 10 Days
Battery</li> <li>✓ Titanium Case</li> <li>✓ Sapphire Crystal</li> <li>✓ Lifetime Warranty</li> </ul> <button
class="pricing-btn">Buy Now</button> </div> </div> </section> </body> </html>
```

Output

TechWatch Pro Logo

Features How It Works Pricing

⌚ Elevate Your Lifestyle

Experience the future of wearable technology with TechWatch Pro

Enter your email address | Get Early Access

Premium Features

❤️ Health Monitoring

Track your heart rate, blood oxygen, sleep quality, and stress levels with medical-grade sensors. Get personalized health insights powered by AI.

🏃 Fitness Tracking

Monitor 100+ workout types with automatic detection. GPS tracking, pace analysis, and real-time coaching to help you reach your fitness goals.

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50-meter water resistance means you can swim, surf, or shower with confidence. Built to withstand any adventure.

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Choose from hundreds of watch faces and interchangeable bands. Express your style with premium materials and finishes.

See It In Action

Choose Your Plan

Starter

$299

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- ✓ Water Resistant
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Buy Now

⭐ Pro

$449

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- ✓ 7 Days Battery
- ✓ GPS + Cellular
- ✓ Premium Bands Included
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Buy Now

Ultra

$599

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- ✓ Unlimited Workouts
- ✓ 10 Days Battery
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Buy Now